TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Advanced multi-material additive manufacturing technology for combining different metallic materials in-process to achieve optimized parts for aerospace systems*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power (√), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Mechanical・Materials		***Mechanical・Materials***	
	Medium Category	Small Category	Medium Category	Small Category
	Metal Materials	*Materials Processing Technologies*	*Aerospace Systems*	*Other Aerospace System Technologies*
Concept of Proposed Technology	*This technology is an approach that uses additive manufacturing processes to combine different metallic materials within a single component and in the same production step. It can be applied across various technologies (powder bed fusion by laser beam, thermal spraying, binder jetting) and makes it possible to place materials with different properties at their optimal locations within the part. This enables the production of components with significantly higher functional integration and performance for key technologies and applications (for example, bimetallic components for space applications, fusion reactors, or high-performance batteries).*			
Rationale for Joint R&D Project	*Fraunhofer IGCV has extensive expertise and the corresponding equipment, enabling, among other things, the reuse of mixed powders in dedicated closed-loop systems in its laboratory. Depending on the material combination, the technology has already reached a high technology readiness level (TRL) and can be transferred into industrial application together with a partner who brings a specific use case. The objective is to industrialize the technology and ensure its successful*			

	industrial application for the partner's component and material combination through targeted know-how transfer.
Key Words	1. *Multi-material additive manufacturing* 2. *Bi-metallic components* 3. *Aerospace systems* 4. *Fusion reactor technology* 5. *Battery technology*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4-5)	End of the Project	Level (7-8)
R&D Objectives	*In the project, the core activity will consist of an extensive process development (parameter variation), which will be evaluated using specific testing procedures. The core objective is the design of the interface of the used two materials and the fitting part design to achieve the desired component performance. The design of the testing parts and the final demonstrators will be finalized by the SME and the research institute. In the final phase, the developed processes will be used to manufacture a demonstrator or component that can be tested in a relevant environment.*			
R&D Target Specifications	*1. The processability of the selected materials is demonstrated (proof of concept) and validated through testing methods (imaging and mechanical).* *2. Through process development and parameter optimization, the selected materials are successfully processed and evaluated using standardized tests (hardness, mechanical strength, porosity).* *3. Manufacturing of a component according to the user's use case (in the case of aerospace: e.g., a bimetallic combustion chamber).*			
R&D Plan	*First, together with the end user, the selection of materials for manufacturing the component (multi-material combustion chamber) will be defined. Subsequently, process parameters will be optimized and the interface between the two materials will be designed. The development work will be continuously accompanied by parallel investigations (hardness testing, metallography, mechanical testing, porosity measurements) to ensure quality. An emphasis will lie on the interface design, where the two materials meet geometrically in the part. Here, mechanical and chemical testing will be conducted to ensure sufficient quality since these areas will define the overall performance of the part.* *The development activities will culminate in the production of an application-relevant component that can be tested in a relevant environment.* *The milestones are:* *1) Demonstration of the processability of the materials.* *2) Demonstration of sufficient quality of the material interface.* *3) Manufacturing of initial demonstrators.* *The required timeline is 36 months.*			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The research institute is responsible for planning and conducting the experiments as well as any necessary hardware modifications. It is also in charge of procuring the raw materials. Furthermore, the research institute is responsible for manufacturing the demonstrators.*
Expected R&R of Korean SME Partner	*The SME partner is expected to provide the use case and to carry out complementary testing activities on the developed samples and components on site. In addition, the partner will provide technical support throughout the overall project. The main activity consists in the design and engineering of the test and demonstrator components in a multi-material configuration, which will serve as the basis for the design trials carried out at the research institute. Also, the final testing of the final manufactured part is placed in the responsibility of the SME partner.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The project will establish advanced multimaterial additive manufacturing capabilities in the space sector, enabling the production of high-performance, next-generation components with optimized material designs. This represents a novel approach that has so far been implemented globally only partially, if at all.* *Market analysts, including the World Economic Forum, estimate that the New Space market will reach USD 4–6 trillion by 2035. The technologies developed in this project will enable Korean SMEs and the broader Korean economy to participate in this market and thereby generate substantial prosperity. In addition, local access to space offers significant geopolitical and logistical advantages.* *Furthermore, the technology can be applied to other key areas, such as the development of fusion reactors and the production of advanced batteries.*

6. References

(1) Brenner, S., Dickmann, M., Helm, R. *et al.* A radio-frequency quadrupole prototype additively manufactured as a multi-material component. *Prog Addit Manuf* **10**, 3951–3961 (2025). https://doi.org/10.1007/s40964-025-01120-6

(2)Schroeder, T., Lehmann, M., Horn, M. *et al.* Transition zone parameter development in multi-material powder bed fusion: a general approach. *Prog Addit Manuf* **9**, 613–624 (2024). https://doi.org/10.1007/s40964-024-00663-4

(3) Thomas Bareth, Daniel Eder, Maja Lehmann, Georg Schlick, Christian Seidel. Multi-material additive manufacturing of conductor-insulator compounds for battery cell cap fabrication. *Materials & Design*, Volume 254, 2025, ISSN 0264-1275, https://doi.org/10.1016/j.matdes.2025.114010.

(4) Christopher Singer, Matthias Schmitt, Georg Schlick, Johannes Schilp. Multi-material additive manufacturing of thermocouples by laser-based powder bed fusion. *Procedia CIRP*, Volume 112, 2022, Pages 346-351, ISSN 2212-8271, https://doi.org/10.1016/j.procir.2022.09.007.

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization